

1. Project Title:

AI-Powered FAQ and Customer Support Automation System.

2. Introduction:

The AI-Powered FAQ and Customer Support Automation System is a web-based application that uses Artificial Intelligence to automatically answer frequently asked questions from customers.

The system allows users to enter their questions through a chat interface. The AI analyses the question and provides a relevant answer from the available FAQ and knowledge database.

If the AI cannot solve the customer's problem, the system can create a support ticket for a human support agent.

3. Project Overview:

The project consists of three major users:

1. Customer

- Asks questions.
- Gets AI-generated answers.
- Creates support tickets.

2. AI Chatbot

- Understands customer questions.
- Searches the FAQ/knowledge base.
- Generates suitable responses.

3. Admin/Support Agent

- Manages FAQs.
- Views customer questions.
- Handles unresolved support tickets.
- Monitors chatbot activity.

4. Key Objectives:

- To develop an AI-based customer support system.

- To automatically answer frequently asked questions.
- To reduce manual customer support work.
- To provide 24/7 automated assistance.
- To maintain customer queries and support tickets.
- To provide an easy-to-use chatbot interface.
- To allow administrators to manage FAQ information.
- To improve response speed and customer experience.

5. Project Instructions:

The system should provide the following features:

Customer Module

- Customer registration/login.
- Chat with AI chatbot.
- Ask questions.
- View previous conversations.
- Create support tickets.
- Check ticket status.

AI Module:

- Receive customer questions.
- Process natural-language questions.
- Search the FAQ/knowledge base.
- Generate appropriate responses.
- Detect unanswered questions.
- Forward difficult queries to support agents.

Admin Module:

- Admin login.
- Add FAQs.
- Update FAQs.

- **Delete FAQs.**
- **View customer conversations.**
- **Manage support tickets.**
- **Assign tickets to support agents.**

6. Steps to Create the Project:

Step 1: Create the Project

Create separate folders for frontend and backend.

```
AI-Customer-Support/  
├── frontend/  
└── backend/
```

Step 2: Develop Frontend

Create:

- Login page
- Registration page
- Chatbot page
- FAQ page
- Ticket page
- Admin dashboard

Step 3: Develop Backend

Create APIs for:

- User authentication
- FAQ management
- Chat messages
- Support tickets
- Admin management

Step 4: Create Database

Use MongoDB to store:

- Users
- FAQs
- Chat history
- Support tickets
- Admin information

Step 5: Integrate AI

Connect the chatbot with an AI service/API or implement a knowledge-based retrieval system.

Step 6: Test the Application

Test:

- Login
- FAQ search
- Chatbot responses
- Ticket creation
- Admin operations
- Database operations

Step 7: Deploy the Project

Deploy the frontend and backend using suitable hosting services.

7. Technologies Used:

- HTML
- CSS
- JavaScript
- React.js
- Node.js
- Express.js

- Mongo DB
- JWT

8. Roles and Responsibilities:

1. Frontend Developer

- Design web pages.
- Create chatbot interface.
- Develop responsive UI.
- Connect frontend with backend APIs.

2. Backend Developer

- Develop REST APIs.
- Implement authentication.
- Handle FAQ and ticket operations.
- Connect backend with MongoDB.

3. AI Developer

- Integrate AI API.
- Develop question-answer processing.
- Connect AI with FAQ/knowledge base.
- Handle unanswered questions.

4. Database Developer

- Design database collections.
- Create relationships between data.
- Maintain data security and consistency.

5. Tester

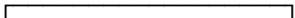
- Test all modules.
- Find bugs.
- Test chatbot responses.
- Check API and database operations.

9. System Architecture:

- The system follows a **three-layer architecture**.

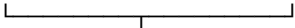
CUSTOMER

|



| FRONTEND |

| React.js |

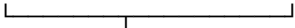


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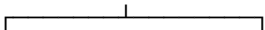


| BACKEND |

| Node.js/Express |



|



| |



| AI/NLP | | MongoDB |

| Engine | | Database |

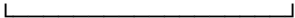


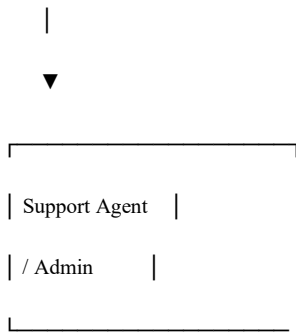
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| FAQ / Knowledge |

| Base |





10.Working Flow:

User asks question



Frontend receives question



Backend API



AI/NLP processing



Search FAQ / Knowledge Base



Answer found?



YES NO



Return Create

Answer Support Ticket



Customer Support Agent

11. Project Working:

The working process is:

START



User Login



Open Chatbot



Enter Question



Backend Receives Question



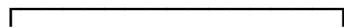
AI Processes Question



Search FAQ / Knowledge Base



Relevant Answer?



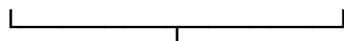
YES

NO



Answer

Create Ticket



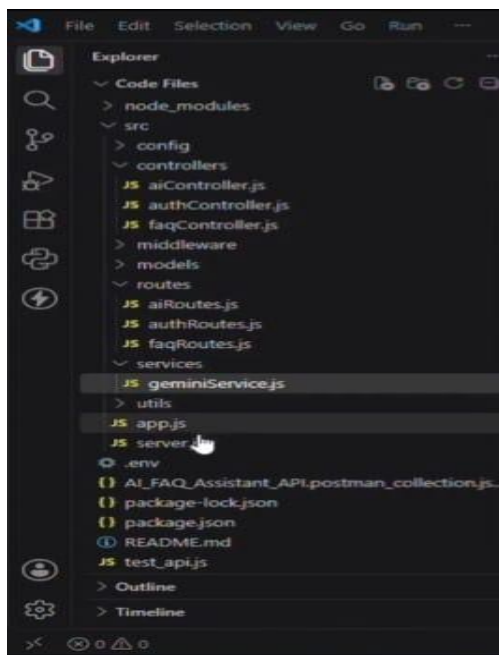


Store Chat



END

12.folder structure:

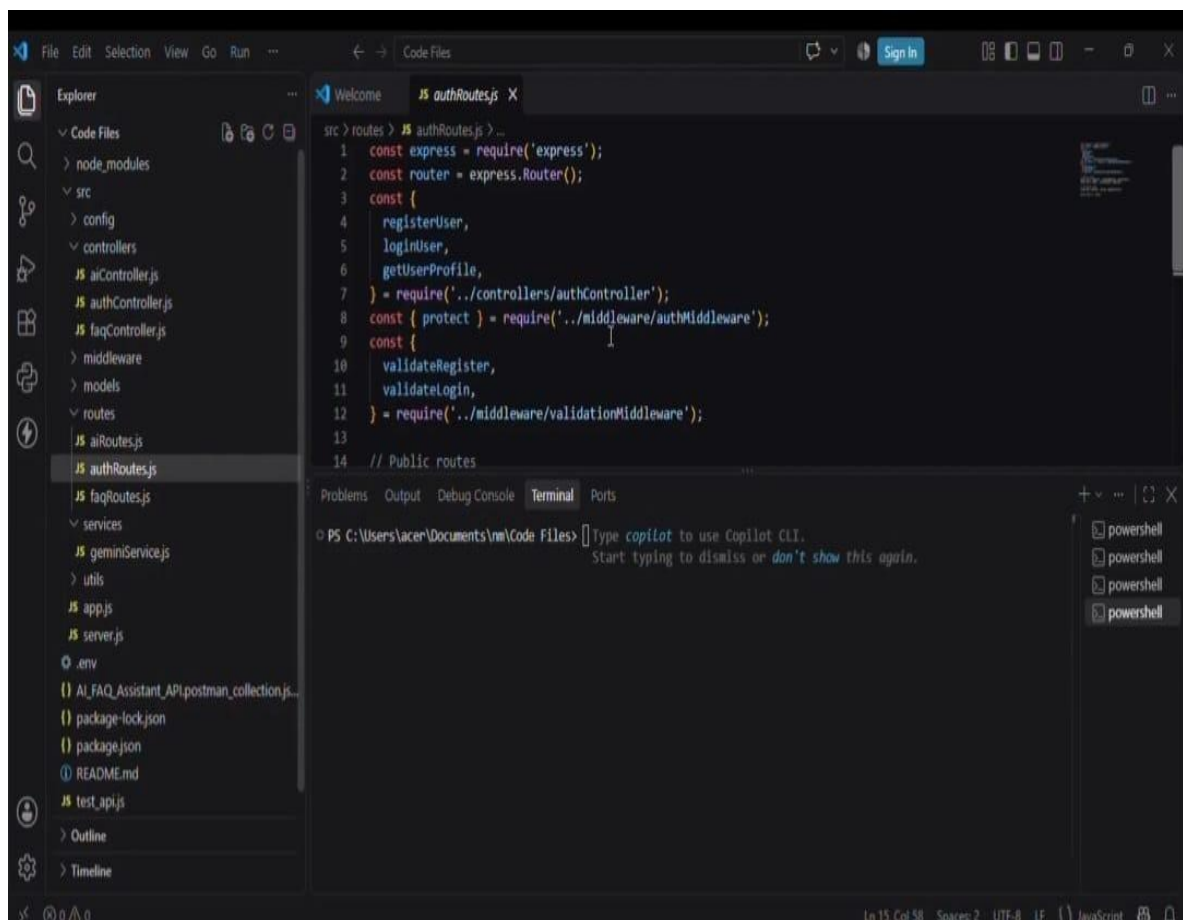


13. Advantages:

1. Provides 24/7 customer support.
2. Reduces repetitive work for support agents.
3. Provides faster responses.

4. Easy to use.
5. Stores customer conversations.
6. Automatically handles common questions.
7. Supports ticket management.
8. Can handle multiple customers.
9. Reduces response time.

14.setup and configuration:



15. Short Project Setup Summary:

Frontend



React.js + HTML + CSS + JavaScript



REST API



Node.js + Express.js



AI/NLP Processing



MongoDB



Users + FAQs + Chats + Tickets

16. Conclusion:

The **AI-Powered FAQ and Customer Support Automation System** provides an efficient way to automate customer service. The system uses an AI chatbot to understand customer questions and provide answers using a FAQ or knowledge base.

When a question cannot be answered automatically, a support ticket can be created for a human agent. The combination of **React.js, Node.js, Express.js, MongoDB, and AI/NLP technology** makes the project suitable as a modern full-stack mini project.